

Nils Matteson

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Inference systems engineer building GPU runtimes, checkpoint/restore, and faster model startup.

Experience

vLLM Project (Inferact-sponsored fellowship) Jul 2026 – Present

Open-Source Fellow

15 authored PRs merged upstream

- Designed and merged **initialized-engine snapshots** using CRIU and CUDA checkpointing, with compatibility checks, failure cleanup, and output validation (#51360). Cut warm-cache activation **63% (54.7 s to 20.4 s)** in H100/Qwen3-32B development benchmarks; validated same-host TP1 capture and two restores across container restart on H200.
- Cut DeepSeek-V4.1-Flash startup to first correct response **28% (731 s to 524 s)** on 2×B300, TP2, with local weights. Reduced FlashInfer MoE autotuning from 266 s to 58 s by narrowing the search; upstream #6032 under review.
- Built vLLM's **pluggable GPU sleep/wake backend** (#44074) and diagnosed NCCL opening InfiniBand in single-GPU workers, blocking CRIU capture. Merged the runtime repair (#59699) and verified two restores on the affected H200 worker.
- Moved Python bytecode compilation into Docker image builds (#55422, merged), saving **15.4 s median per pair** from local-registry pull to first correct response in three paired Qwen3-0.6B/RTX 4090 Laptop tests, for 1.8% larger compressed images.
- Extended snapshot preparation/recovery into ordinary `vllm serve` (#59851, draft). Validated original-server recovery and two restored copies on one GPU; failed recovery keeps public serving unavailable.

thaw (open-source inference infrastructure; Matteson Systems LLC)

Apr 2026 – Present

Founder & Engineer

Rust · CUDA · Python

- Built a **Rust/CUDA snapshot engine** for LLM weights and KV/prefix-cache state. A prewarmed H100/Llama-3.1-8B pool completed four branches (64-token limit) in **0.88 s median per round** across five rounds after 22.3 s one-time initialization.
- Restored a 16 GB Llama-3-8B weight snapshot from cold-cache local NVMe at **13.0 GB/s** on H100 using double-buffered `0_DIRECT` reads and pinned-memory DMA. Validated bit-identical 70B output through weight snapshot and sleep/wake on 2×H100.

University of Wisconsin–Madison, Research Cyberinfrastructure

Jan 2026 – Apr 2026

AI Workflows Research Assistant

AWS Bedrock · RAG · LLM evaluation

- Built an AWS Bedrock evaluation and cost framework comparing **9 LLMs and 3 ensembles on 282 questions**; presented *Deploying RAG in Bedrock vs. Local* at UW's ML+X Forum.

Selected Research

Re-feeding Is Not Replaying: Measuring Replay Noise in Counterfactual Token-Credit Estimation

Jun 2026

Sole author · arXiv:2606.15621 (preprint)

- Designed a three-pass experiment across **3,434 token pivots and 3 models**; re-feeding prompts caused **14–28 percentage points** of excess disagreement in low-margin token-credit estimates above replica noise. Batch-invariant kernels eliminated the measured disagreement; released code and per-pivot data.

Education

Northeastern University – M.S. Computer Science, in progress

Expected 2028

University of Wisconsin–Madison – B.S. Data Science, Minor in Computer Science

May 2026

Technical Skills

Languages: Python, Rust, C++, CUDA

Systems: vLLM, PyTorch, SGLang, CUDA graphs, NCCL, CRIU/cuda-checkpoint, Linux, Docker, Buildkite